DR. LIAM MOROY

AI researcher specialised in generative models and Bayesian methods, applying ML research to large-scale real-world problems.

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SUMMARY

Research Associate at Heriot-Watt University (BISC group), working on deep generative models, primarily diffusion-based, for ill-posed inverse problems with robust and uncertainty-aware reconstruction as the goal. Experienced across the full research cycle, from problem formulation and algorithm development to large-scale GPU experimentation, publications, and open-source code. Currently building quantitative-finance expertise through a generative-modelling project on implied volatility surfaces.

EDUCATION

Ph.D. – ONERA, DTIS/MIC group

- Université Paris-Saclay.
- Generative AI (diffusion models) for solving inverse problems in imaging, applied to tomographic imaging.

2022 - 2025   Paris, France

MSc Electrical Engineering + MSc Complex Systems

- Enseirb-Matmeca engineering school, Bordeaux INP and University of Bordeaux (concurrent).

2020 - 2022   Bordeaux, France

BSc - Electronics Engineering

- Enseirb-Matmeca engineering school, Bordeaux INP.

2017 - 2020   Bordeaux, France

RESEARCH ACTIVITIES

Selected Publications:

- "On the Posterior Gap in Plug&Play Diffusion Methods for Sparse-View CT", IEEE JSTSP 2026 Special Issue on High-Dimensional Imaging: Emerging Challenges and Advances in Reconstruction and Restoration, L. Moroy, G. Bourmaud, F. Champagnat, J.-F. Giovannelli.
- "Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT", ICASSP 2025, L. Moroy, G. Bourmaud, F. Champagnat, J.-F. Giovannelli.

Selected Oral Presentations:

- "Improving the Posterior Sampling of PnP Diffusion Models by Super-Calibration: application to Poisson inverse problem" 7th SLSIP Workshop, May 2026, Krujë, Albania.
- "Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT", GdR IASIS: Advances in Learning-Based Image Restoration, Dec 2024, Henri Poincaré Institute, Paris.

TEACHING & COMMUNICATION

University of Bordeaux

- Master ISC – MSc teaching assistant - Introduction to noise and random signals (2023/24 & 2024/25).

ENSEIRB-MATMECA engineering school

- Engineering cycle – MSc teaching assistant - Generative modelling with diffusion models (2024/25).
- Engineering cycle – Freshman teaching assistant - Introduction to programming in C/C++ (2023/24 & 2024/25).

Interactive scientific outreach

- Interactive in-browser demos of generative modelling methods (annealed Langevin sampling, diffusion posterior sampling for inpainting), aimed at technical and non-specialist audiences: liammry.github.io.

PROFESSIONAL EXPERIENCE

Research Associate – Generative AI for Inverse Problems in Physical Sciences

Heriot-Watt University (BISC)

Dec. 2025 - Present | Edinburgh, Scotland

- Development and application of deep generative models (diffusion models) as learned priors for solving highly ill-posed inverse problems across multiple physical imaging modalities: Computed Tomography for nuclear applications, non-conventional optics, and medical imaging.
- Collaboration within a multi-disciplinary team of statisticians, physicists, and engineers on high-impact scientific and medical imaging applications.

Python | Pytorch/CUDA | Generative AI | Bayesian Inversion

R&D Internship – *Deep Learning* for tomographic reconstruction of compressible flows

ONERA (DTIS/MIC)

Feb. 2022 - Aug. 2022 | Paris, France

- Design and evaluation of learned prior models for Bayesian CT reconstruction of compressible flows from limited measurements.

Python | Pytorch/CUDA | Generative AI | Bayesian Inversion

R&D Internship – *Deep Learning* for nebula removal in astronomical images

Astrophysics Laboratory of Bordeaux (LAB) & IMS Laboratory

Jul. 2021 - Sep. 2021 | Bordeaux, France

- Training a regressor from noisy ground truths and transfer learning from synthetic datasets.

Python | Pytorch/CUDA | C/C++ | Image Processing

SELECTED PROJECTS

Generative Modelling of Implied Volatility Surfaces

Independent project – github.com/LiamMry/iv-surface-diffusion

2026

- Score-based diffusion model (VP-SDE, attention U-Net) trained on $\sim$3,500 daily SPY implied-volatility surfaces (2010–2023); no-arbitrage consistency emerges from data without explicit constraints.
- Frames daily surface calibration as posterior sampling under a diffusion prior; benchmarked ODE/SDE samplers (Euler, Heun, DPM-Solver).

Quantitative Finance | Diffusion Models | PyTorch

SKILLS

Programming
- Python, C, C++

ML & Scientific Libraries
- PyTorch / CUDA
- NumPy, Matplotlib

Tools & Workflow
- Git / GitHub
- LaTeX
- Linux / Bash
- Slurm / HPC clusters

Research Expertise
- Diffusion / score-based generative models
- Plug-&-Play methods
- Bayesian inference & inverse problems
- Statistical data analysis & experimental design
- Uncertainty quantification
- Signal & image processing

Languages
- French (native)
- English (TOEIC 950 – fluent)
- Spanish (B1 – intermediate)
- Japanese (beginner)